

Functional Annotation of PP_2483 (UniProt Q88K10) in *Pseudomonas putida* KT2440

1. Summary — Answer to the Research Question

PP_2483 (Q88K10) is a MobA-superfamily nucleotidyltransferase, specifically a MocA-type CTP:molybdopterin cytidyltransferase (EC 2.7.7.76). Its primary function is to catalyze the Mg^{2+} -dependent, CTP-specific transfer of a cytidyl (CMP) group onto the C4' terminal phosphate of molybdopterin (Mo-MPT), producing the **molybdopterin cytosine dinucleotide (MCD)** form of the molybdenum cofactor plus pyrophosphate. It acts in the **cytoplasm** as a biosynthetic (cofactor-maturation) enzyme. Genomic context pinpoints its physiological purpose: PP_2483 lies in a self-contained gene cluster (PP_2477–PP_2483) that also encodes an **isoquinoline 1-oxidoreductase** (PP_2477/PP_2478; a xanthine-oxidase-family molybdo-hydroxylase experimentally shown to carry the MCD cofactor), an **XdhC-type MCD-insertion chaperone** (PP_2480), and the Moco-biosynthesis enzyme **MoaA** (PP_2482) — so PP_2483 supplies the MCD cofactor that activates this local molybdo-hydroxylase, contributing to N-heterocyclic aromatic compound catabolism. The identification rests on domain architecture, orthology (KEGG K07141), paralog discrimination, this operonic context with a genuine MCD enzyme, and homology to biochemically characterized *E. coli* MocA.

⚠ **Identity/evidence note:** Q88K10 is a TrEMBL (computationally annotated, unreviewed) entry and there is **no primary experimental study of PP_2483 itself**. The functional call is an *inference* from orthology, conserved domains, genomic context, and experimentally characterized homologs (chiefly *E. coli* MocA). It should be regarded as a high-confidence prediction, not a directly demonstrated activity.

2. Protein at a Glance

Property	Value
UniProt	Q88K10 (Q88K10_PSEPK), unreviewed /TrEMBL
Gene / locus	PP_2483 (ordered locus name); GenBank AAN68095; RefSeq WP_010953419.1
Organism	<i>Pseudomonas putida</i> KT2440 (taxid 160488)
Length / mass	199 aa / 21.2 kDa
Genomic position	complement(2,830,343..2,830,942)
Domain	MobA-like NTP transferase (Pfam PF12804, res. 12–171)
Fold	Nucleotide-diphospho-sugar transferase / SpsA-like (SUPFAM SSF53448; Gene3D 3.90.550.10; CDD cd04182 "GT_2_like_f")
InterPro	IPR025877 (MobA-like NTP transferase); IPR029044 (Nucleotide-diphospho-sugar transferases superfamily)
Orthology	KEGG K07141 = molybdenum cofactor cytidyltransferase, EC 2.7.7.76 ; eggNOG COG2068
GO	GO:0016779 (nucleotidyltransferase activity)
Cofactor	Mg ²⁺ (UniProt "Magnesium" keyword)
Predicted localization	Cytoplasm (no signal peptide / transmembrane segment)

3. Key Findings and Evidence

3.1 The protein is a Mg²⁺-dependent nucleotidyltransferase of the MobA-like superfamily

- The single structural domain (res. 12–171) is a MobA-like NTP transferase domain (Pfam PF12804), belonging to the nucleotide-diphospho-sugar transferase / SpsA-like fold

(SSF53448, Gene3D 3.90.550.10). The GO annotation is nucleotidyltransferase activity (GO:0016779).

- The N-terminus carries the canonical nucleotidyltransferase phosphate-binding motif (... G-L-V-L-A-A-G-T-G-Q-R...), which in this superfamily binds the α -phosphate of the incoming NTP and coordinates a catalytic Mg^{2+} — consistent with the UniProt Magnesium ligand keyword.
- **Bioinformatic/evolutionary evidence.** This superfamily (MobA, MocA, GlnU, RmlA, IspD, sugar-1-P nucleotidyltransferases) shares a common fold and mechanism: NTP + acceptor-phosphate $\rightarrow$ NDP/NMP-acceptor + PPi.

3.2 Orthology and paralog discrimination pin the substrate to CTP + molybdopterin (MocA, EC 2.7.7.76), not GTP (MobA, EC 2.7.7.77)

- KEGG assigns PP_2483 to **K07141 (molybdenum cofactor cytidyltransferase, EC 2.7.7.76)**.
- Critically, the *guanylyl*-transferring paralog **MobA (K03752, EC 2.7.7.77)** is encoded by a **separate** KT2440 gene, **PP_3457**. Because the two activities are catalyzed by distinct, nucleotide-specific paralogs (~22 % identity), the presence of a dedicated MobA elsewhere in the genome supports PP_2483 being the **cytidyl** (MocA) branch (a second K07141 paralog, PP_4230, also exists).
- In the characterized *E. coli* system, MocA is strictly CTP-specific (ATP and GTP are not used) and requires Mg^{2+} , and the two paralogs differ functionally by two sub-domains: the **N-terminal domain dictates nucleotide (C vs G) specificity**, while the **C-terminal domain dictates which acceptor molybdoenzyme is bound** (Neumann et al., 2009,

P 19542235

Neumann et al., 2011,

P 21081498

3.3 Reaction catalyzed

$$\text{CTP} + \text{Mo-molybdopterin} \xrightarrow{\text{Mg}^{2+}} \text{PP}_{2483} \text{ (MocA)} + \text{molybdopterin cytosine dinucleotide (MCD)}$$
- Substrate specificity: donor = **CTP** (specific); acceptor = **Mo-MPT** (molybdenum-loaded molybdopterin); essential metal = **Mg^{2+}** . (By homology to *E. coli* MocA: $K_d \approx 0.23 \mu\text{M}$ for CTP, $\approx 1.2 \mu\text{M}$ for Mo-MPT; $k_{\text{cat}} \approx 0.37 \text{ min}^{-1}$.)

3.4 Genomic context identifies the specific downstream molybdoenzyme: an isoquinoline 1-oxidoreductase

PP_2483 sits at the end of a complete, self-contained molybdo-hydroxylase gene cluster (PP_2477–PP_2483) that encodes both a xanthine-oxidase-family enzyme **and** its dedicated cofactor-biosynthesis/maturation machinery:

Gene	Product	Role
PP_2477	Isoquinoline 1-oxidoreductase α subunit (K07302, EC 1.3.99.16; Pfam Fer2/Fer2_2)	[2Fe-2S] electron-transfer subunit
PP_2478	Isoquinoline 1-oxidoreductase β subunit (K07303, EC 1.3.99.16; Pfam MoCoBD + Ald_Xan_dh_C)	Molybdopterin (MCD)-binding catalytic subunit
PP_2480	XdhC-family accessory factor (Pfam XdhC_C, XdhC_CoxI)	MCD sulfuration + insertion chaperone
PP_2482	MoaA / GTP 3',8-cyclase (K03639, EC 4.1.99.22)	First committed step of Moco biosynthesis
PP_2483	MocA cytidyltransferase (this gene)	Synthesizes the MCD cofactor

Isoquinoline 1-oxidoreductase (IOR) is an experimentally established MCD enzyme. Purified IOR from *Pseudomonas* / *Brevundimonas diminuta* is a molybdenum hydroxylase containing **1 mol CMP per enzyme**, molybdenum, and two [2Fe-2S] clusters, with the CMP originating from a **molybdopterin cytosine dinucleotide (MCD)** cofactor (Lehmann et al., 1994,

[P 8157655](#));
Canne et al., 1997,

[P 9245410](#));
Israel et al., 2002,

[P 12023088](#))).
XdhC-type chaperones (as encoded by PP_2480) are specifically required to sulfurate and insert MCD into xanthine-oxidase-family apoenzymes (Neumann & Leimkühler, 2011,

[P 21151514](#))).

Therefore the biological role of PP_2483 is to synthesize the MCD cofactor for the co-operonic isoquinoline 1-oxidoreductase (PP_2477/PP_2478), a molybdo-hydroxylase that hydroxylates isoquinoline to 1-oxo-1,2-dihydroisoquinoline — linking PP_2483 to catabolism of N-heterocyclic aromatic compounds. This dedicated co-localization with a genuine MCD enzyme is decisive evidence for the cytidylyl (MCD/MocA) assignment over the guanylyl (MobA) alternative.

Confirming the acceptor-specific "one MocA per molybdoenzyme operon" architecture.

KT2440 encodes a second MocA/K07141 paralog, **PP_4230**, which lies in its *own* independent molybdo-hydroxylase operon: PP_4231 (XdhC accessory factor, K07402), PP_4232 (cytochrome c), PP_4233 ([2Fe-2S] small oxidoreductase subunit), and **PP_4234 (aldehyde oxidase / xanthine dehydrogenase, large MoCoBD subunit)**. Thus each of the two *Pseudomonas* MocA paralogs is co-encoded with a distinct xanthine-oxidase-family molybdoenzyme and its own XdhC chaperone. This parallel genomic architecture is direct evidence for **dedicated, acceptor-specific MCD delivery** — matching the biochemical finding that the MocA C-terminal domain determines binding to its specific acceptor protein (Neumann et al., 2011,

P 21081498

— and pins **PP_2483 to the isoquinoline 1-oxidoreductase** while PP_4230 serves the aldehyde oxidase/xanthine dehydrogenase.

3.4b Caveat — nucleotide specificity is NOT assignable from global sequence identity

Needleman-Wunsch global identity: PP_2483 is actually closer to *E. coli* **MobA** (34.3%) than to *E. coli* **MocA** (25.3%), while *E. coli* MobA vs MocA share only 24.2%, and PP_2483 vs its paralog PP_4230 = 48.1%. Because C- vs G-selectivity is dictated by a few N-terminal active-site residues rather than by overall identity (Neumann et al., 2011,

P 21081498

), the higher global similarity to MobA does **not** imply guanylyl specificity. The cytidylyl (MCD) assignment rests on orthology (KEGG K07141) **plus** the functional/genomic context above, not on sequence identity — this is an inference and should be experimentally confirmed.

3.5 Localization

- Molybdenum-cofactor and MCD biosynthesis are cytoplasmic processes; the 199-aa protein has no predicted signal peptide or transmembrane helix, consistent with a **soluble cytoplasmic** enzyme. (Inference; no experimental localization data exist for PP_2483.)

4. Pathway Context

- **Pathway:** Molybdenum cofactor (Moco) biosynthesis → molybdopterin *dinucleotide* branch. Upstream: MoaA→MoaC→MoaD/MoaE (cPMP → MPT), MogA/MoeA (Mo insertion → Mo-MPT). PP_2483 (MocA) then adds CMP to Mo-MPT to yield **MCD**, which is delivered by an XdhC-type chaperone to apo-molybdoenzymes of the xanthine oxidase family. (KEGG maps this branch under ppu00790 "folate biosynthesis," the map that also houses molybdopterin biosynthesis.)
 - **Downstream physiology:** the MCD product is inserted (via XdhC PP_2480) into the co-encoded isoquinoline 1-oxidoreductase (PP_2477/PP_2478), which hydroxylates isoquinoline to 1-oxo-1,2-dihydroisoquinoline (EC 1.3.99.16). This links PP_2483 to catabolism of N-heterocyclic aromatic compounds in this metabolically versatile soil bacterium. A second MocA paralog (PP_4230) likely serves a different molybdoenzyme.
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5. Supported and Refuted Hypotheses

Hypothesis	Verdict	Basis
Member of MobA-like NTP-transferase superfamily; Mg ²⁺ -dependent nucleotidyltransferase	Supported	Pfam PF12804, SSF53448, GO:0016779, Mg keyword, N-terminal motif
Specifically a CTP:molybdopterin cytidyltransferase (MocA, EC 2.7.7.76) producing MCD	Supported (high-confidence inference)	KEGG K07141; distinct MobA gene (PP_3457) present; operonic with a genuine MCD enzyme (isoquinoline 1-oxidoreductase) + XdhC chaperone
PP_2483 supplies MCD specifically to the co-encoded isoquinoline 1-oxidoreductase (PP_2477/PP_2478)	Supported (strong inference)	Operon structure; IOR shown to contain MCD (P 8157655 / 9245410/12023088); XdhC PP_2480 present
Cytidyl specificity can be read from global sequence identity to MobA/MocA	Refuted	PP_2483 is closer to <i>E. coli</i> MobA (34%) than MocA (25%); specificity is set by local residues, not global identity
It is the MobA guanylyltransferase (EC 2.7.7.77)	Refuted	MobA/K03752 is a separate gene (PP_3457)
It is a sugar-1-phosphate nucleotidyltransferase (NDP-sugar biosynthesis)	Refuted / unlikely	Orthology + Moco/molybdoenzyme genomic context favor MCD biosynthesis over sugar-nucleotide metabolism
Cytoplasmic, soluble enzyme	Supported (inference)	No signal/TM; Moco biosynthesis is cytoplasmic

6. Limitations and Future Directions

- **No direct experimental data** exist for PP_2483; the annotation is inferred from orthology, conserved domains, genomic context, and characterized homologs. Q88K10 is an unreviewed TrEMBL entry.
 - KEGG K07141 is an automated ortholog assignment; two KT2440 paralogs (PP_2483, PP_4230) carry it. Operon structure assigns PP_2483 to the isoquinoline 1-oxidoreductase (PP_2477/PP_2478) and PP_4230 to the aldehyde oxidase/xanthine dehydrogenase (PP_4233/PP_4234), but the acceptor specificity of each MocA (set by its C-terminal domain) has not been experimentally verified in *P. putida*.
 - The isoquinoline 1-oxidoreductase MCD evidence derives from orthologous enzymes in *Pseudomonas/Brevundimonas diminuta*; the KT2440 IOR (PP_2477/PP_2478) itself has not been directly shown to carry MCD, though its MoCoBD/Ald_Xan_dh_C domain architecture and operon are diagnostic.
 - **Recommended experiments:** (i) in-vitro reconstitution (MocA + Mo-MPT + CTP + Mg^{2+} → MCD) to confirm cytidyltransferase activity and CTP specificity; (ii) ΔPP_2483 phenotyping for loss of isoquinoline 1-oxidoreductase activity / isoquinoline degradation; (iii) AlphaFold model + active-site residue mapping against MobA/MocA structures to verify the cytosine-selecting N-terminal motif; (iv) confirm the dedicated PP_2483→IOR versus PP_4230→aldehyde oxidase acceptor pairing (e.g., cross-complementation).
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7. Key References

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